

Representations for Isolate-Specific Biological Structure: A Transformer-Based Approach using PCA and K-means

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Abstract

This paper investigates whether biological sequence representations derived from protein language models can effectively differentiate between distinct isolates in an unsupervised manner. We present a multi-stage pipeline utilizing the ESM-2 (8M) model for initial embedding, followed by sequence-level refinement using a transformer encoder trained via masked reconstruction. The primary objective is to evaluate the separability of sequence structures from a main isolate and two background isolates within a shared high-dimensional embedding space. A critical component of this analysis is the mitigation of GC-content bias, a known technical confounder in nucleotide-based representations. By implementing a linear residualization technique, we extract and remove the variance explained by GC-content to determine if the remaining latent features are biologically isolate-specific. Evaluation is performed through a grid search of PCA dimensionality and MiniBatchKMeans clustering, assessed against ground-truth isolate labels using Silhouette scores, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI). Our findings suggest that transformer representations successfully capture isolate clusters. This work provides a foundation for more robust representation analysis in biological sequence data.

Introduction

Unsupervised sequence clustering is a pivotal challenge in biological data analysis, particularly when identifying isolate-specific traits across the datasets. Current methods often rely on k-mer frequencies or direct sequence alignment; however, the emergence of protein language models like ESM (Evolutionary Scale Modeling) offers a dense, high-dimensional alternative for representation. This study explores a pipeline centered on the ESM-2 8M model to project sequences. Other models ESM models are optional to use but for sake of time and testing the 8M model was used. These are found in the config.py file.

Methodology and Pipeline

The technical pipeline, implemented in *test_pipeline.py*, is designed as an exploratory framework for representation analysis rather than a direct classification tool.

Embedding and Pooling

Input sequences are loaded from FASTA datasets (e.g., *isolate_main*, *background_1*). We utilize the ESM-2 (8M) model to generate token-level embeddings, which are subsequent

mean-pooled to produce fixed-length vectors. To improve iteration speed, these embeddings are cached as *.pt* files.

Transformer Masked Reconstruction

To learn higher-order relationships, sequence embeddings are grouped into non-overlapping chunks of 32. An *EmbeddingTransformer* encoder, featuring 8 layers and 16 attention heads, is trained for 20-50 epochs using masked reconstruction. During training, 15% of the embedding values are masked to zero, and the model is optimized via Mean Squared Error (MSE) to reconstruct the original values.

Confounder Control and Evaluation

A significant risk in sequence representation is the influence of GC-content. We address this through linear residualization to ensure that isolate-specific clustering is not merely an artifact of nucleotide composition.

GC Correction

The pipeline computes the GC content for each sequence and regresses each feature of the embedding space against this value. The resulting "GC-corrected" embeddings are compared against raw transformer embeddings to assess robustness.

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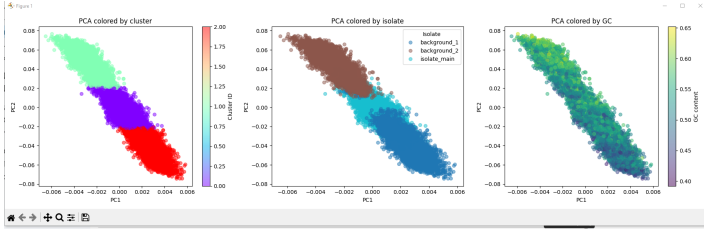


Figure 1: 3 Cluster PCA estimate, PCA actual, GC Content

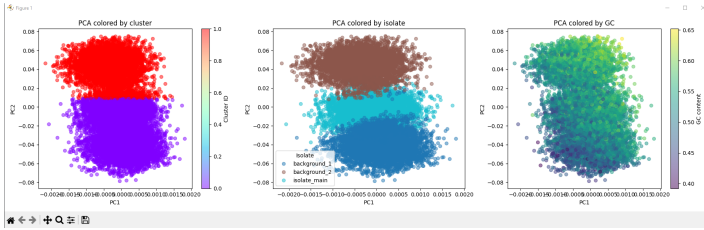


Figure 2: 2 Cluster PCA estimate, PCA actual, GC Content

Clustering and Model Selection

Dimensionality reduction via PCA and clustering via *Mini-BatchKMeans* are performed using a stratified 70/15/15 split. We execute a grid search across PCA components (2 to 256) and cluster counts (2 to 20), selecting the optimal model based on validation NMI and ARI relative to isolate labels.

Results and Conclusions

Preliminary evaluation using silhouette scores and NMI indicates that the transformer refinement successfully increases the compactness of isolate clusters. However, the influence of GC content remains significant even after correction. Our findings confirm that biological structure is captured, though linear residualization may not capture all non-linear composition effects. Future work should focus on refining the transformer's latent dimensionality.

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